1. When a probability distribution is symmetrical about the mean, the skewness is equal to zero. If the distribution has a longer tail to the right than to the left, the skewness is positive, and we say that the distribution is skewed to the right. On the contrary, if the distribution has a longer tail to the left than to the right, the skewness is negative, and we say that the distribution is skewed to the left. For a random variable X with mean μ and variance σ^2 , the measure of skewness is defined as

$$\text{Skewness} = \frac{E[(X-\mu)^3]}{\sqrt[3]{\tau^2}}$$

The probability mass function of a discrete random variable X is given by

$$p_X(X=k) = \begin{cases} 2^{-k}, & k = 1,2 \\ \frac{1}{4}, & k = 3 \end{cases}$$

(a) [15 pts.] Find the mean and variance of X.
(b) [10 pts.] Is the distribution skewed to the right or left?

2. Let X be a random variable of the MIXED TYPE having the following cumulative distribution function.

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - \frac{2}{3}e^{-x} & x \geq 0 \end{cases}$$

(a) [5 pts.] Carefully sketch the graph of $F_X(x)$.

(b) [10 pts.] Find the mean and variance of X.
Hint: $\int xe^{-x}dx = -e^{-x}(x+1)$, $\int x^2e^{-x}dx = -e^{-x}(x^2+2x+2)$

(c) [10 pts.] Find $P[X = 0]$, $P[0 < X < 1]$, and $P[0 \leq X \leq 1]$.

3. Suppose X and Y are zero mean independent Gaussian random variables with common variance 1. Define

$$\boxed{\begin{aligned} R &= \sqrt{X^2 + Y^2}, & 0 \le R < \infty \\ \theta &= \text{Arctan}\left(\frac{Y}{X}\right), & -\pi < \theta < \pi \end{aligned}}$$

(a) [10 pts.] What is the joint probability density function $p_{X,Y}(x, y)$ of X and Y?

(b) [15 pts.] Obtain the joint probability density function $p_{R,\theta}(r, \theta)$ of R and θ.

Hint: First, find X and Y in terms of R and θ. Then, obtain the Jacobian matrix of the transformation.

4. The joint probability density function $p_{X,Y}(x,y)$ of continuous random variables X and Y is given by

$$\boxed{p_{X,Y}(x,y) = \begin{cases} 2, & 0 < x < y < 1 \\ 0, & \text{otherwise} \end{cases}}$$

(a) [5 pts.] Indicate the region in the two-dimensional $X - Y$ space where $p_{X,Y}(x,y)$ is non-zero.

(b) [10 pts.] Find $p_{Y|X}(y|X = x)$, the conditional probability density function of Y given $X = x$.

(c) [10 pts.] Compute, $E_{Y|X}[Y|X = x]$, the conditional expected value Y given $X = x$.

Marmara University
Engineering Faculty
Electrical and Electronics Engineering
STAT2056 - Probability and Random Variables
Final Exam - Solutions

1. (a)

k	1	2	3
$P[X=k]$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

$E[X] = 1\cdot\left(\frac{1}{2}\right)+2\cdot\left(\frac{1}{4}\right)+3\cdot\left(\frac{1}{4}\right) = \frac{2+2+3}{4} = \frac{7}{4}$ (5)

$E[X^2] = 1^2\cdot\left(\frac{1}{2}\right)+2^2\cdot\left(\frac{1}{4}\right)+3^2\cdot\left(\frac{1}{4}\right) = \frac{2+4+9}{4} = \frac{15}{4}$ (5)

$Var(X) = E[X^2]-(E[X^2])^2 = \frac{15}{4}-\left(\frac{7}{4}\right)^2 = \frac{60-49}{16} = \frac{11}{16}$ (5)

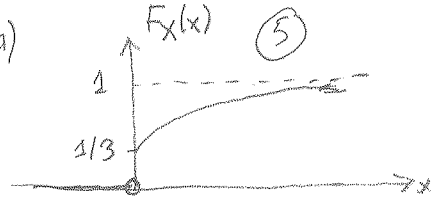
(b) $E[(X-\mu)^3] = E\left[\left(X-\frac{7}{4}\right)^3\right] = \left(1-\frac{7}{4}\right)^3\cdot\frac{1}{2}+\left(2-\frac{7}{4}\right)^3\cdot\frac{1}{4}+\left(3-\frac{7}{4}\right)^3\cdot\frac{1}{4}$

$= -\frac{27}{64}\cdot\frac{1}{2}+\frac{1}{64}\cdot\frac{1}{4}+\frac{125}{64}\cdot\frac{1}{4} = \frac{-54+1+125}{256} = \frac{72}{256}$ (4)

$Skewness = \frac{72/256}{\sqrt[3]{11/16}} > 0$ (4) $\Rightarrow$ Distribution is skewed to the right. (2)

---

2. (a)



(5)

(b) $P_X(x) = \frac{dF_X(x)}{dx} = \frac{1}{3}\delta(x)+\frac{2}{3}e^{-x}u(x)$ (2)

$E[X] = \int_{-\infty}^{\infty} x\cdot P_X(x)\,dx = \int_{-\infty}^{\infty} x\cdot\left[\frac{1}{3}\delta(x)+\frac{2}{3}e^{-x}u(x)\right]dx$ (3)

$= \frac{1}{3}\int_{-\infty}^{\infty} x\delta(x)\,dx + \frac{2}{3}\int_0^{\infty} x\,e^{-x}dx = \frac{2}{3}\left(-e^{-x}(x+1)\Big|_0^{\infty}\right) = \frac{2}{3}$

$E[X^2] = \int_{-\infty}^{\infty} x^2\cdot P_X(x)\,dx = \int_{-\infty}^{\infty} x^2\cdot\left[\frac{1}{3}\delta(x)+\frac{2}{3}e^{-x}u(x)\right]dx = \frac{1}{3}\int_{-\infty}^{\infty} x^2\delta(x)\,dx + \frac{2}{3}\int_0^{\infty} x^2\cdot e^{-x}dx$

$= \frac{2}{3}\left[-e^{-x}(x^2+2x+2)\Big|_0^{\infty}\right] = \frac{2}{3}\cdot 2 = \frac{4}{3}$ (3)

$Var(X) = E[X^2]-(E[X])^2 = \frac{4}{3}-\left(\frac{2}{3}\right)^2 = \frac{4}{3}-\frac{4}{9} = \frac{8}{9}$ (2)

(c) $P[X=0] = \frac{1}{3}$ (The height of jump at $X=0$ in the graph of $F_X(x)$) (4)

$P[0<X<1] = \int_0^1 \frac{2}{3}e^{-x}dx = -\frac{2}{3}e^{-x}\Big|_0^1 = \frac{2}{3}(1-e^{-1})$ "Impulse at 0 is not included" (4)

$P[0\le X\le 1] = \int_0^1\left(\frac{1}{3}\delta(x)+\frac{2}{3}e^{-x}\right)dx = \frac{1}{3}+\frac{2}{3}(1-e^{-1})$ "Impulse at 0 is included" (2)

---

3. (a) (5)

$P_{X,Y}(x,y) = P_X(x)\cdot P_Y(y) = \frac{1}{\sqrt{2\pi}}\cdot e^{-\frac{x^2}{2}}\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{y^2}{2}} = \frac{1}{2\pi}e^{-\frac{x^2+y^2}{2}}$ (5)

(b) $X = R\cos(\theta)$, $y = R\sin(\theta)$ (5)

$J = \begin{bmatrix}\frac{\partial x}{\partial R} & \frac{\partial x}{\partial\theta}\\ \frac{\partial y}{\partial R} & \frac{\partial y}{\partial\theta}\end{bmatrix} = \begin{bmatrix}\cos(\theta) & -R\sin(\theta)\\ \sin(\theta) & R\cos(\theta)\end{bmatrix}$ (3)

$|\det(J)| = |R\cos^2(\theta)+R\sin^2(\theta)| = R$ (2)

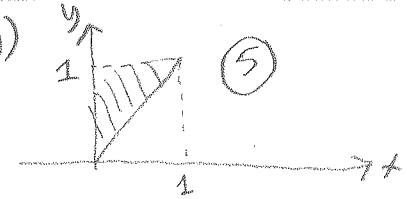
$P_{R,\theta}(r,\theta) = |\det(J)|\cdot P_{X,Y}(R\cos(\theta), R\sin(\theta))$

$= r\cdot\frac{1}{2\pi}e^{-\frac{1}{2}(r^2\cos^2(\theta)+r^2\sin^2(\theta))}$ (5)

$= \frac{r}{2\pi}e^{-\frac{r^2}{2}}$, $0<r<\infty$, $-\pi<\theta<\pi$

---

4. (a)



(5)

(b) $P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)}$ (5)

$P_X(x) = \int_{-\infty}^{\infty} P_{X,Y}(x,y)\,dy = \int_x^1 2\,dy = 2y\Big|_x^1 = 2(1-x)$

$P_{Y|X}(y|x) = \frac{2}{2(1-x)} = \frac{1}{1-x}$ (5)

(c) $E_{Y|X}[Y|X=x] = \int_{-\infty}^{\infty} y\cdot P_{Y|X}(y|x)\,dy$ (5)

$= \int_x^1 y\cdot\frac{1}{1-x}\,dy = \frac{1}{1-x}\cdot\frac{y^2}{2}\Big|_x^1$

$= \frac{1}{1-x}(1-x^2) = \frac{x+1}{2}$ (5)